Agostinho A. Soto-wang

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SUMMARY

An aspiring immuno-oncology researcher, invested in the growth of one's community and self through education. Passionate about implementing machine learning/data algorithms to explain biological phenomena.

EDUCATION

UC San Diego
B.S, Bioengineering: Bioinformatics Expected June 2029

- 3.882 GPA

EXPERIENCE

Mechanotransduction in Immune Network Dynamics Lab, La Jolla, CA **01/2026-Current**
Volunteer Lab Research Assistant

- Developed python scripts and macros for facilitating data visualization and formatting
- Created computer vision segmenting model for brightfield microscopy using OpenCV
- Assisting in developing predictive tensorflow classification models for cell-spheroid interactions
- Developed quantitative protocol for analyzing collagen alignment using open-source software

UCSD Extension Learning Machine Learning Course, La Jolla, CA **10/2024-06/2025**
Futures Scholarship

- Created self-implementation of K Nearest Neighbors, Logistic Regression, Gradient Descent of Loss function
- Utilized SKLearn, Pandas, NumPy Libraries
- Developed an object oriented deep neural network (multi-layer perceptron)
- Interpreting and manipulating large datasets, visualizing key trends

Projects/Research

Immune Cell Infiltration Prediction **01/2026 - Present**

- Explored high dimensional data of immune cell morphology and dynamics
- Developed logistic regression and working on Long Short-Term Memory Model for predicting cell infiltration
- Utilizing layer-wise relevance propagation and SHAP to explain important features and timepoints

Computer Vision Model for Protrusion **04/2026 – Present**

- Trained YOLO computer vision model to identify cancer protrusions from brightfield imaging
- Develop train dataset by marking experimental data
- Working towards quantifying protrusion length, straightness, and aggression

Cell Migration Analysis Across Conditions **05/2026 – Present**

- Analyze single-cell migration data over time
- Utilize MSD, autocorrelation function, singular value decomposition to characterize cell patterns

SKILLS & ACTIVITIES

- Python: Tensorflow, Pandas, Numpy, Matplotlib
- Fluent English and Mandarin, beginning Spanish
- Confocal Image Processing: FIJI
- Biomedical Engineering Society: Lab Expo Lead
- Electric Guitar and Running